
Same-Class Batch Normalization Statistics Are Worse Than Wrong-Class: A Forensic Study of an Asymmetric Substitution Effect

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Abstract

1 Replacing the global batch-normalization (BN) running statistics of a trained con-
2 volutional network with statistics estimated only from samples of one class is a
3 benign-sounding intervention—class-conditional statistics are, after all, just the
4 natural per-class refinement of the global ones. We show this intervention has a
5 sharp and counter-intuitive structure. On CIFAR-10 with a SmallResNet, substi-
6 tuting class- c statistics when evaluating class- c samples reduces accuracy from
7 91.1% to 3.5%, while substituting a deliberately mismatched class’s statistics
8 leaves 65.4% accuracy. The same-class condition is thus roughly $26\times$ more dam-
9 aging than the wrong-class one, and a simple “out-of-distribution input” explana-
10 tion cannot account for the asymmetry. The ordering replicates across four archi-
11 tectures (SmallResNet, VGG-11-BN, SimpleCNN, WideResNet-28-10) and three
12 datasets (CIFAR-10, CIFAR-100, Tiny-ImageNet-200). Two additional observa-
13 tions rule out the most natural alternative explanations: (i) a linear probe on the
14 post-substitution penultimate features achieves $\geq 99.7\%$ accuracy on CIFAR-10
15 (and 100% on the larger-label-space datasets), so the representation itself is not
16 damaged; (ii) GroupNorm models, which have no running statistics to swap, are
17 unaffected; and (iii) temperature scaling reduces the resulting expected calibration
18 error from 0.394 to 0.151 but recovers exactly 0% of the lost accuracy, so the fail-
19 ure is not a logit-scale issue. A linear interpolation between class-conditional and
20 global statistics yields a smooth monotone recovery in classifier accuracy but a
21 non-monotone dip in probe accuracy near $\alpha \approx 0.6$. We treat these findings as a
22 forensic observation about BN-statistic substitution, and discuss interpretations.

23 1 Introduction

24 Batch normalization [Ioffe and Szegedy, 2015] stores per-layer running mean and variance vectors
25 that are used at inference to standardize activations. The running statistics are typically thought of
26 as a benign approximation to the true population mean and variance: a numerical convenience that
27 affects optimization but not what the network represents.

28 We test that picture by perturbing the running statistics in a controlled way. For a network trained on
29 C classes, we estimate one set of running statistics per class by passing the corresponding subset of
30 the training data through the network in BN `train()` mode without updating any parameters. We
31 then run inference under four conditions: *global* (the standard buffers), *same-class* (class- c statistics
32 for class- c samples), *wrong-class* (class- $(c + \lfloor C/2 \rfloor) \bmod C$ statistics for class- c samples), and
33 *random-class* (a random other class’s statistics for each sample).

A skeptical reading of any single one of these conditions is straightforward: the classifier was trained against the global statistics, so swapping in any other statistics is a distribution shift, and accuracy drops are unsurprising. What this paper documents is a pattern that the distribution-shift reading does not predict. Same-class statistics are not a mild refinement of global statistics; they are the *worst* of the three substitutions, and by a wide margin. On CIFAR-10 with a SmallResNet, same-class accuracy is 3.5%—below the 10% chance baseline—while wrong-class accuracy is 65.4% and random-class accuracy is 59.0%. The ordering “same $\ll$ random $<$ wrong $\ll$ global” replicates across four architectures and three datasets.

We document this asymmetry as the headline contribution and pair it with three controls that constrain its interpretation. (1) The penultimate features remain class-discriminative: a linear probe trained on the post-substitution features reaches $\geq 99.7\%$ accuracy on CIFAR-10 and 100% on CIFAR-100 and Tiny-ImageNet-200, exceeding the global baseline. (2) GroupNorm [Wu and He, 2018] variants of the same architecture, which carry no running statistics, are unaffected by construction. (3) Temperature scaling [Guo et al., 2017] reduces the same-class expected calibration error from 0.394 to 0.151 but recovers exactly 0% of the lost accuracy: the failure is in the logit *ranking*, not its scale. A continuous interpolation between class-conditional and global statistics shows a smooth monotone recovery in classifier accuracy, but the linear probe is non-monotone with a minimum near $\alpha \approx 0.6$.

Contributions.

1. **An asymmetric substitution ordering.** Same-class BN statistics are roughly $26\times$ more damaging than wrong-class on CIFAR-10 (3.5% vs 65.4%) and the ordering same $\ll$ random/wrong $\ll$ global replicates across four architectures (SmallResNet, VGG-11-BN, SimpleCNN, WideResNet-28-10) and three datasets (CIFAR-10, CIFAR-100, Tiny-ImageNet-200).
2. **The features survive.** A linear probe on the post-substitution penultimate features achieves 99.9% accuracy on CIFAR-10 (above the global baseline of 90.9%), and 100% on CIFAR-100 and Tiny-ImageNet-200. The substitution does not damage the representation.
3. **Two ruled-out alternative explanations.** The effect requires BN’s running-statistic mechanism (GroupNorm models retain 81.6% accuracy) and is not a logit-scale problem (temperature scaling halves ECE but recovers 0% accuracy).
4. **A non-monotone interpolation.** Linearly interpolating from class-conditional to global statistics produces a smooth monotone recovery of classifier accuracy but a non-monotone probe-accuracy curve, indicating that the two endpoints correspond to qualitatively different geometric regimes for the activation space.

We treat the paper as a forensic empirical study of a specific intervention, not a sweeping mechanistic claim about what BN statistics “are”; Section 5 discusses interpretations.

2 Background and Related Work

Batch normalization. BN [Ioffe and Szegedy, 2015] normalizes activations using mini-batch statistics during training and exponentially-averaged *running* statistics $(\mu_\ell, \sigma_\ell^2)$ at inference, after which a learned affine transform γ_ℓ, β_ℓ is applied. Santurkar et al. [2018] and Bjorck et al. [2018] argue BN’s main effect is to smooth the optimization landscape rather than to address “internal co-variate shift”. These works examine BN’s training-time role; we examine the inference-time role of the running buffers specifically.

BN statistics as adaptable quantities. A number of methods treat BN statistics as something that can be re-estimated rather than treated as fixed. AdaBN [Li et al., 2018] replaces them with target-domain statistics for domain adaptation; TENT [Wang et al., 2021] and Schneider et al. [2020] update them on test data for robustness to distribution shift. Frankle et al. [2020] show that training only the BN affine parameters of a randomly-initialized network can reach surprisingly high accuracy. These works modify BN parameters in directions that *help*; we instead document a regime in which a seemingly benign modification is sharply harmful.

84 **Linear probing.** Linear probes [Alain and Bengio, 2016, Kornblith et al., 2019] measure whether
 85 class structure is linearly accessible in a representation. We use them in a counterfactual role: a
 86 network that mispredicts but whose features still admit a high-accuracy linear probe has a problem
 87 in the head, not in the encoder.

88 **Calibration.** Guo et al. [2017] introduced expected calibration error (ECE) and temperature scal-
 89 ing. We use both as diagnostics. Temperature scaling is monotone in logits and so cannot change the
 90 argmax; if it nonetheless recovered accuracy, the same-class failure would be a mere over-confidence
 91 problem, which it is not.

92 **Normalization variants.** GroupNorm [Wu and He, 2018] computes statistics over feature-map
 93 groups within each example and stores no running averages; we use it as an architectural control.
 94 Normalization-free architectures [Brock et al., 2021] replace BN entirely.

95 3 Methodology

96 **Architectures.** We train four image classifiers on CIFAR-10 with standard augmentation (random
 97 crop pad 4, horizontal flip), SGD with momentum 0.9, weight decay 5×10^{-4} , initial LR 0.1, cosine
 98 annealing. **SmallResNet:** a four-stage residual net with widths [32, 64, 128, 256] and 20 BN layers
 99 (10 seeds, 20 epochs); a five-stage **SmallResNet64** variant with widths [32, 64, 128, 256, 512] is
 100 used for 64×64 inputs. **VGG-11-BN** (8 BN layers, 3 seeds), **SimpleCNN** (4-layer plain CNN,
 101 4 BN layers, 3 seeds), **WideResNet-28-10** [Zagoruyko and Komodakis, 2016] (depth 28, widen
 102 factor 10, dropout $p=0.3$, 3 seeds). We additionally train SmallResNet on CIFAR-100 (5 seeds, 25
 103 epochs) and SmallResNet64 on Tiny-ImageNet-200 (3 seeds, 25 epochs). Full hyperparameters are
 104 in Appendix A.

105 **Class-conditional statistics.** Let f_θ be a trained CNN with BN layers $\{B_\ell\}$ carrying running
 106 means μ_ℓ and running variances σ_ℓ^2 . For class c we reset the BN buffers, place the model in `train()`
 107 mode (so BN updates the buffers via its standard EMA), iterate in `torch.no_grad()` over the class-
 108 c training subset, and snapshot the resulting buffers as $(\mu_\ell^{(c)}, (\sigma_\ell^{(c)})^2)$. *No model parameters are*
 109 *updated.* The original buffers are restored before the next class’s pass.

110 **Inference conditions.** For test sample x with label c we evaluate four conditions: *global* (load
 111 original buffers), *same-class* (load $(\mu_\ell^{(c)}, (\sigma_\ell^{(c)})^2)$), *wrong-class* (load class- $(c + \lfloor C/2 \rfloor) \bmod C$),
 112 and *random-class* (load a uniformly random class $\neq c$, drawn per-sample). For each condition we
 113 record top-1 accuracy, mean true-class softmax confidence, and the penultimate-layer features.

114 **Linear probe.** Penultimate features (post global average pooling) are split 80/20; we fit ridge re-
 115 gression ($\lambda=10^{-3}$, one-hot targets, closed form) for CIFAR-10, and multinomial logistic regression
 116 (L-BFGS, $C=1.0$) for ≥ 100 -class settings, and report held-out accuracy.

117 **Interpolation.** For $\alpha \in \{0.0, 0.1, \dots, 1.0\}$ we set $\mu_\alpha = \alpha\mu_{\text{global}} + (1 - \alpha)\mu_{\text{same}}$ and $\sigma_\alpha^2 =$
 118 $\alpha\sigma_{\text{global}}^2 + (1 - \alpha)\sigma_{\text{same}}^2$ and evaluate.

119 **Calibration controls.** ECE uses 15 equal-width bins. Temperature scaling [Guo et al., 2017]
 120 optimizes a single scalar T on a held-out validation set under NLL, then divides logits by T . Because
 121 $T > 0$ preserves the argmax, any accuracy change under temperature scaling is by construction zero;
 122 the diagnostic value lies in comparing ECE before and after.

123 **GroupNorm control.** We retrain SmallResNet with all BatchNorm layers replaced by Group-
 124 Norm (8 groups, fewer when channels are smaller). GroupNorm has no running statistics, so the
 125 substitution paradigm is inapplicable and the relevant question is whether the architecture suffers
 126 any analogous loss.

127 **Statistical protocol.** SmallResNet/CIFAR-10 uses 10 seeds (42–51); SmallResNet/CIFAR-100
 128 uses 5 (42–46); all other configurations use 3 (42–44). We report mean $\pm$ std and paired t -tests
 129 where stated.

Table 1: SmallResNet on CIFAR-10 across the four BN-statistic conditions. Mean $\pm$ std over 10 seeds.

Condition	Accuracy (%)	Confidence (%)	Linear Probe (%)
Global (baseline)	91.08 $\pm$ 0.12	88.88 $\pm$ 0.15	90.86 $\pm$ 0.55
Same-class	3.55 $\pm$ 0.69	4.80 $\pm$ 0.52	99.88 $\pm$ 0.09
Wrong-class	65.37 $\pm$ 4.69	50.64 $\pm$ 3.22	100.00 $\pm$ 0.00
Random-class	58.99 $\pm$ 5.84	45.47 $\pm$ 5.19	97.63 $\pm$ 1.11

Table 2: Cross-architecture results on CIFAR-10 (mean $\pm$ std). The ratio same/wrong is < 0.18 in every architecture.

Architecture	BN layers	Global	Same-class	Wrong-class	Same/Wrong
SmallResNet ($n = 10$)	20	91.08 $\pm$ 0.12	3.55 $\pm$ 0.69	65.37 $\pm$ 4.69	0.054
VGG-11-BN ($n = 3$)	8	85.44 $\pm$ 1.28	10.88 $\pm$ 1.91	62.96 $\pm$ 1.25	0.173
SimpleCNN ($n = 3$)	4	83.17 $\pm$ 0.23	26.33 $\pm$ 0.48	78.10 $\pm$ 0.46	0.337
WRN-28-10 ($n = 3$)	many	90.61 $\pm$ 0.51	11.12 $\pm$ 0.90	64.06 $\pm$ 3.14	0.174

4 Results

4.1 The asymmetric substitution ordering

Table 1 shows the four-condition comparison on our most-replicated configuration. Three observations follow.

First, every non-global condition lies strictly below global accuracy, as a generic distribution-shift account would predict. Second, the ordering among the three substitution conditions is the opposite of what such an account would predict: same-class statistics, which by any natural metric are *closest* to global (they differ from global only by within-class variation, not between-class variation), are the most damaging. Same-class accuracy (3.5%) lies below the 10% chance baseline; wrong- and random-class accuracies (65.4% and 59.0%) sit much higher. Same-vs-global accuracy is $t(9) = -374.1, p = 3.5 \times 10^{-20}$; same-vs-wrong is $t(9) = -39.1, p < 10^{-10}$.

Third, per-class deltas are uniformly large and negative across all 10 CIFAR-10 classes (Appendix B), ranging from -0.97 for *automobile* to -0.72 for *cat*; the collapse is not driven by one or two pathological classes.

Cross-architecture replication. Table 2 shows the ordering replicates across all four architectures we tested. In every case, same-class accuracy is well below wrong-class accuracy, and the ratio is at least $4\times$ for the BN-heavy networks.

The same-class accuracy correlates roughly with the number of BN layers: SmallResNet (20 layers) drops the most, SimpleCNN (4 layers) the least. WideResNet-28-10 (a deep BN-heavy network with dropout $p=0.3$) shows a same-class accuracy of 11.1%, somewhat higher than SmallResNet’s 3.5% despite many more BN layers; the dropout regularization may be partially buffering the head against activation perturbation.

Cross-dataset replication. Table 3 extends the experiment to CIFAR-100 and Tiny-ImageNet-200 with SmallResNet/SmallResNet64. The accuracy collapse is even sharper in larger label spaces: same-class accuracy is below the chance baseline ($1/C$) on both datasets. With $C = 100$ or $C = 200$, all three substitution conditions fall to floor and the same-vs-wrong gap is no longer measurable, but the substitution effect itself is more severe.

4.2 The features survive: a probe-vs-classifier dissociation

The right-most columns of Tables 1 and 3 report the linear probe accuracy under same-class statistics. Despite classification accuracy of 3.5% on CIFAR-10 and 0.17% on Tiny-ImageNet-200, a linear probe trained on the same penultimate features achieves 99.9% and 100.0% respectively—in the latter case, with zero variance across seeds.

Table 3: Cross-dataset results with SmallResNet (CIFAR-10/100) and SmallResNet64 (Tiny-ImageNet-200). Chance is $1/C$. Same-class accuracy lies *below* chance on both larger datasets.

Dataset	C	Chance	Global	Same-class	Wrong-class	Probe (same)
CIFAR-10	10	10.00%	91.08 ± 0.12	3.55 ± 0.69	65.37 ± 4.69	99.88 ± 0.09
CIFAR-100	100	1.00%	70.11 ± 0.57	0.60 ± 0.49	1.00 ± 0.00	100.00 ± 0.00
Tiny-ImageNet-200	200	0.50%	60.77 ± 0.30	0.17 ± 0.24	0.50 ± 0.00	100.00 ± 0.00

Table 4: Temperature scaling on the same-class condition (SmallResNet, CIFAR-10, $n=3$ seeds). Temperature scaling is by construction monotone in logits and cannot change the argmax; the relevant comparison is the ECE before/after.

Metric	Before ($T=1$)	After ($T \approx 3.43$)
Accuracy (%)	2.87 ± 0.49	2.87 ± 0.49
ECE	0.394 ± 0.013	0.151 ± 0.003

162 This dissociation matters for interpreting Section 4.1: whatever the same-class substitution is doing,
 163 it is not destroying the encoder’s class-discriminative information. A linear refit of the head on
 164 the same features is essentially perfect. The classifier’s inability to recover the class is therefore a
 165 property of the *specific learned head*, not of the representation it operates on. Notably the probe
 166 accuracy under same-class often *exceeds* the global-statistic probe (99.9% vs 90.9% on CIFAR-10),
 167 suggesting that class-conditional normalization removes within-class variation in a way that makes
 168 the resulting features more linearly separable.

169 The same-class probe is somewhat lower for SimpleCNN (54.7%; see Appendix C), the smallest
 170 architecture; on the BN-heavy networks (SmallResNet, VGG, WRN) the probe sits at $\geq 98.4\%$.

171 4.3 Two ruled-out explanations

172 **Generic normalization-swap effect.** A natural alternative explanation is that the asymmetry is an
 173 artifact of *any* deep architecture with running statistics that are perturbed at inference. We test this
 174 by training SmallResNet with all BatchNorm layers replaced by GroupNorm [Wu and He, 2018],
 175 which computes statistics from the current input and has no running buffers (3 seeds, 20 epochs).
 176 The GroupNorm model achieves $81.65 \pm 1.94\%$ accuracy and a $81.08 \pm 1.64\%$ linear probe under
 177 standard inference; there are no running statistics to substitute, so the asymmetry is inapplicable to
 178 it by construction. The effect is thus specific to the running-statistic mechanism in BN.

179 **Logit-calibration issue.** A second alternative is that same-class statistics merely sharpen or
 180 rescale the logits in a way that softmax confidence and ECE pick up but argmax does not. We
 181 test this with temperature scaling [Guo et al., 2017]: optimize a single scalar T on validation NLL
 182 under same-class statistics, then divide all logits by T at test time.

183 The optimum is $T \approx 3.43$: same-class logits are substantially over-scaled. Temperature scaling
 184 cuts ECE from 0.394 to 0.151—a substantial calibration improvement—but recovers 0% of the
 185 lost accuracy. This is by construction (monotone-in-logits transforms preserve argmax), but the
 186 diagnostic content is the residual ECE: even after the optimal scalar rescaling the probabilities are
 187 still poorly calibrated. The under-the-hood logit *ranking* has been disrupted, not just its scale. The
 188 same-class collapse is therefore not a temperature problem.

189 4.4 A non-monotone interpolation between class-conditional and global

190 We linearly interpolate the BN buffers between same-class ($\alpha = 0$) and global ($\alpha = 1$) and evaluate
 191 at each step. Table 5 shows four representative points; the full 11-point sweep is in Appendix E.

192 Two features are notable. First, classifier accuracy rises smoothly and monotonically from 3.4% to
 193 91.0%—there is no phase transition. Second, the linear probe is *non-monotone*: it is highest at the
 194 same-class endpoint (99.9%), declines to a minimum of 74.65% at $\alpha = 0.6$, and partially recovers
 195 to 90.85% at the global endpoint. Near $\alpha \approx 0.6$ –0.7 the two curves cross.

Table 5: Linear interpolation from same-class ($\alpha = 0$) to global ($\alpha = 1$) statistics on SmallResNet/CIFAR-10. Classifier accuracy rises monotonically; the linear probe is non-monotone, with a minimum near $\alpha \approx 0.6$. The full 11-point sweep is in Appendix E.

α	Accuracy (%)	Probe (%)	
0.0	3.44	99.90	class-conditional endpoint
0.3	23.14	94.85	
0.6	65.17	74.65	probe minimum
1.0	91.01	90.85	global endpoint

The non-monotone probe trajectory is constraining for any explanation of the same-class effect. If the substitution simply moved features uniformly off-distribution, one might expect probe accuracy to be a monotone function of α (whether increasing or decreasing). Instead, the two endpoints both admit high-quality linear separability while an intermediate point does not, suggesting that the activation space at $\alpha = 0$ and $\alpha = 1$ correspond to two qualitatively different geometric regimes, with intermediate statistics partially mixing the two.

5 Discussion

The four findings of Section 4 together constrain how the same-class effect can be explained. Section 4.3 rules out two of the most natural alternatives: it is not a generic property of running-statistic perturbation (GroupNorm has no running statistics and is unaffected), and it is not a logit-scale issue (temperature scaling halves ECE without recovering accuracy). What remains is the question of why the *ordering* is what it is.

An OOD-input reading is not enough. A bare distribution-shift reading—“the classifier was trained against global statistics, anything else is OOD”—would predict that all three substitution conditions hurt similarly, since all three move the activation distribution off the training distribution. The data is incompatible with this prediction in a robust way: same-class is consistently the most damaging condition (Tables 1, 2), and on CIFAR-10 it is more damaging than wrong-class by more than $18\times$ in the SmallResNet case. Whatever same-class statistics are doing, they are doing it more thoroughly than a generic OOD perturbation.

An interpretation: BN statistics as an implicit calibrator. One interpretation consistent with the data is that the trained classification head depends on a tight alignment between the running statistics and its learned weights, and that same-class statistics shift the activation distribution in the direction that the head is most sensitive to for that class—because those statistics encode the class-mean activation pattern. Wrong-class and random-class statistics shift it in directions less specifically aligned with the head’s class- c template. Linear refitting (the probe) absorbs whatever offset the substituted statistics induce, which is why probe accuracy is high in every condition. In this reading, BN running statistics function as an output *calibrator* that couples the encoder to the head, and same-class statistics maximally decouple them. We emphasize that this is one possible interpretation rather than a proven mechanism; alternatives—e.g., that same-class normalization specifically collapses the between-class signal that the head exploits—are also consistent with the observations and may be complementary.

The non-monotone probe. Under any of these interpretations, the non-monotone probe (Section 4.4) is informative: it indicates that the same-class endpoint is not just “OOD” relative to global, but a qualitatively different organization of the activation space (one in which within-class variation is suppressed and the features are very linearly separable). The intermediate regime mixes the two and is worse for linear separability than either endpoint.

Connection to prior empirical work. The interpretation also gives a perspective on why methods like AdaBN [Li et al., 2018] and TENT [Wang et al., 2021] work: they re-estimate BN statistics on data the classifier head must work with, recalibrating the encoder-to-head coupling rather than learning new features. Our results predict that their benefit is largely orthogonal to representation quality, and that pairing them with classifier-head adjustment should magnify their effect. The same

lens applies to several other practical settings: BN-folding at inference fuses the running statistics into the convolution, making the calibrator-head coupling an architectural property rather than a runtime one; normalization-free designs such as NFNets [?] avoid the coupling entirely; and post-hoc model compression that quantizes or prunes BN buffers should be evaluated for the kind of asymmetric sensitivity we document, since seemingly small statistic perturbations can have outsized effects on accuracy without affecting the underlying representation.

Methodological implication. A network at 0.17% accuracy on Tiny-ImageNet-200 may have a 100%-linearly-separable representation. Reading classification accuracy as a proxy for representation quality is unsafe in any setting in which the encoder-to-head alignment can be perturbed independently of the encoder—domain shift, adversarial inputs, normalization-layer interventions. Diagnosing such failures benefits from separately measuring representation quality (e.g., via a linear probe) and head-alignment (e.g., via ECE plus a temperature-scaling check).

6 Limitations and Conclusion

Limitations. (i) All experiments are on natural-image classification at scales up to Tiny-ImageNet-200. Whether the asymmetry holds on ImageNet-1k, fine-grained recognition, or non-vision domains is open. (ii) Architectures with non-BN normalization (LayerNorm, RMSNorm, transformers) carry no running statistics and are not directly addressed. (iii) Wrong-class is operationalized via the fixed mapping $c \mapsto (c + \lfloor C/2 \rfloor) \bmod C$; characterizing the full $C \times C$ surface of stat-class pairings would be a richer analysis. (iv) We replace all BN layers simultaneously; layer-wise replacement could reveal which depths are most sensitive. (v) When we replace running statistics we leave the affine parameters $(\gamma_\ell, \beta_\ell)$ unchanged; cleanly disentangling their contribution is left to future work. (vi) ECE with fixed binning has known sensitivities [Nixon et al., 2019]; we use 15 bins following standard practice.

Conclusion. Substituting class-conditional BN running statistics at inference produces a sharp asymmetric ordering—same $\ll$ random/wrong $\ll$ global—that holds across four architectures and three datasets. The encoder remains class-discriminative under the same-class substitution (linear probe $\geq 99.7\%$), and the failure is neither a generic normalization-swap effect (GroupNorm models are unaffected) nor a logit-scale issue (temperature scaling halves ECE but recovers 0% accuracy). A linear interpolation between class-conditional and global statistics shows a smooth monotone recovery of classifier accuracy and a non-monotone probe minimum near $\alpha \approx 0.6$. We treat this as a forensic observation about a specific intervention rather than a sweeping claim about the role of BN statistics, and we discuss interpretations consistent with it. The data we report constrain—rather than determine—a mechanistic story; we hope the asymmetry itself becomes a useful target for theoretical work on BN.

Reproducibility. Code and data are available at <https://anonymous.4open.science/r/bn-calibration>.

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303 A Experimental configuration

Table 6: Full experimental configuration.

Parameter	Value
SmallResNet	BasicBlock, [2, 2, 2, 2], channels [32, 64, 128, 256], 20 BN layers
SmallResNet64	BasicBlock, 5-stage, channels [32, 64, 128, 256, 512], for 64×64
VGG-11-BN	Standard VGG-11 with BN, 8 BN layers
SimpleCNN	4-layer CNN, 4 BN layers
WideResNet-28-10	depth=28, widen_factor=10, dropout=0.3
Training epochs	20 (CIFAR-10), 25 (CIFAR-100, Tiny-ImageNet-200)
Batch size	128
Optimizer	SGD (momentum=0.9, weight_decay= 5×10^{-4})
Learning rate	0.1, cosine annealing
Augmentation	RandomCrop(32, pad=4), RandomHorizontalFlip (CIFAR)
Normalization (input)	$\mu=(0.4914, 0.4822, 0.4465)$, $\sigma=(0.2470, 0.2435, 0.2616)$
Linear probe	Ridge ($\lambda=10^{-3}$, CIFAR-10) or LogisticRegression (≥ 100 -class), 80/20 split
ECE	15 equal-width bins
Temperature scaling	NLL-optimized scalar T on validation set
Seeds	42–51 (SR/CIFAR-10), 42–46 (SR/CIFAR-100), 42–44 (others)
Device	Apple Silicon (MPS backend)

304 B Per-class accuracy deltas

305 On CIFAR-10/SmallResNet, per-class same-minus-global accuracy deltas (averaged over 10 seeds)
 306 range from -0.97 (*automobile*) to -0.72 (*car*). All ten classes show deltas ≤ -0.72 , ruling out the
 307 hypothesis that the collapse is concentrated in a small subset of classes.

308 C Cross-architecture results in detail

Table 7: Cross-architecture results on CIFAR-10 with all four conditions and the linear probe (mean $\pm$ std, $n=3$ seeds for VGG/SimpleCNN/WRN).

Architecture	Condition	Acc. (%)	Conf. (%)	Probe (%)
VGG-11-BN	Global	85.44 ± 1.28	80.47 ± 1.78	85.00 ± 1.80
	Same-class	10.88 ± 1.91	10.29 ± 1.23	98.37 ± 0.63
	Wrong-class	62.96 ± 1.25	58.61 ± 1.31	99.57 ± 0.27
	Random-class	57.50 ± 3.03	52.29 ± 3.12	95.28 ± 0.78
SimpleCNN	Global	83.17 ± 0.23	75.68 ± 0.11	80.53 ± 1.44
	Same-class	26.33 ± 0.48	20.71 ± 0.18	54.68 ± 0.74
	Wrong-class	78.10 ± 0.46	65.79 ± 0.59	93.67 ± 0.27
	Random-class	72.11 ± 0.36	59.20 ± 0.33	90.82 ± 0.52
WRN-28-10	Global	90.61 ± 0.51	89.36 ± 0.43	91.02 ± 0.28
	Same-class	11.12 ± 0.90	11.04 ± 0.73	99.78 ± 0.09
	Wrong-class	64.06 ± 3.14	52.64 ± 3.51	99.60 ± 0.07
	Random-class	62.41 ± 1.76	51.21 ± 2.25	96.63 ± 0.44

309 D Cross-dataset results in detail

310 E Full interpolation sweep

311 F Detailed ECE and temperature-scaling sweep

312 Note that wrong-class statistics produce moderate ECE (0.064) despite substantially reduced accu-
 313 racy: the network’s confidence and accuracy remain roughly aligned, just at a lower level. Same-

Table 8: Full four-condition results on CIFAR-100 (SmallResNet, $n=5$ seeds, logistic-regression probe).

Condition	Acc. (%)	Conf. (%)	Probe (%)
Global	70.11 ± 0.57	63.98 ± 0.73	64.23 ± 1.34
Same-class	0.60 ± 0.49	0.94 ± 0.00	100.00 ± 0.00
Wrong-class	1.00 ± 0.00	1.00 ± 0.00	100.00 ± 0.00
Random-class	1.00 ± 0.63	1.00 ± 0.00	65.49 ± 8.75

Table 9: Full four-condition results on Tiny-ImageNet-200 (SmallResNet64, $n=3$ seeds, logistic-regression probe).

Condition	Acc. (%)	Conf. (%)	Probe (%)
Global	60.77 ± 0.30	53.75 ± 0.24	51.83 ± 0.55
Same-class	0.17 ± 0.24	0.49 ± 0.00	100.00 ± 0.00
Wrong-class	0.50 ± 0.00	0.50 ± 0.00	100.00 ± 0.00
Random-class	0.50 ± 0.00	0.50 ± 0.00	57.02 ± 3.41

class statistics produce severely miscalibrated outputs (ECE 0.394) and accuracy near floor. The optimal temperature $T \approx 3.43$ indicates the same-class logits are substantially over-scaled.

G Per-seed tables

Table 10: 11-point interpolation from same-class ($\alpha = 0$) to global ($\alpha = 1$) statistics on SmallResNet/CIFAR-10.

α	Accuracy (%)	Linear Probe (%)
0.0	3.44	99.90
0.1	7.13	99.70
0.2	13.41	98.67
0.3	23.14	94.85
0.4	36.47	87.10
0.5	51.61	77.60
0.6	65.17	74.65
0.7	75.26	77.88
0.8	82.62	83.00
0.9	87.64	87.53
1.0	91.01	90.85

Table 11: Expected Calibration Error by condition (SmallResNet, CIFAR-10, $n=3$ seeds).

Condition	Accuracy (%)	ECE
Global (baseline)	91.33 ± 0.12	0.025 ± 0.001
Same-class	2.87 ± 0.49	0.394 ± 0.013
Wrong-class	63.40 ± 4.26	0.064 ± 0.013

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Table 12: Per-seed temperature-scaling on the same-class condition.

Seed	Glob. ECE	SC ECE	SC ECE (post- T)	T
42	0.026	0.382	0.150	3.456
43	0.024	0.412	0.148	3.428
44	0.026	0.388	0.154	3.416

Table 13: SmallResNet CIFAR-10 – Classification accuracy by seed.

Seed	Global	Same-Class	Wrong-Class	Random-Class
42	90.96	3.85	60.19	55.26
43	91.05	4.02	70.70	67.73
44	91.01	2.46	69.93	55.45
45	91.07	4.06	64.59	56.74
46	91.26	3.84	57.08	56.07
47	91.30	3.37	60.49	57.01
48	90.95	2.39	67.19	50.82
49	91.19	2.90	63.57	56.01
50	90.93	4.22	69.00	66.09
51	91.04	4.35	70.98	68.72

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Table 14: SmallResNet CIFAR-10 – Linear probe accuracy by seed.

Seed	Global	Same-Class	Wrong-Class	Random-Class
42	90.30	99.90	100.00	96.30
43	90.65	99.80	100.00	96.85
44	91.10	99.90	100.00	97.05
45	89.65	99.90	100.00	96.90
46	90.80	100.00	100.00	98.30
47	91.80	99.95	100.00	99.35
48	90.80	99.90	100.00	96.80
49	91.15	99.90	100.00	98.90
50	91.00	99.65	100.00	96.65
51	91.30	99.90	100.00	99.15

Table 15: SmallResNet CIFAR-100 – Per-seed results ($n=5$).

Seed	Glob. Acc	SC Acc	Glob. Probe	SC Probe
42	70.95	1.00	65.65	100.00
43	69.68	1.00	64.85	100.00
44	70.21	0.00	65.00	100.00
45	69.31	1.00	61.80	100.00
46	70.42	0.00	63.85	100.00

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Table 16: WideResNet-28-10 CIFAR-10 – Per-seed accuracy ($n=3$).

Seed	Global	Same-Class	Wrong-Class	Random-Class
42	90.17	10.44	64.21	62.49
43	90.35	12.40	67.83	64.53
44	91.32	10.53	60.15	60.22

Table 17: WideResNet-28-10 CIFAR-10 – Per-seed linear probe ($n=3$).

Seed	Global	Same-Class	Wrong-Class	Random-Class
42	91.40	99.85	99.70	96.30
43	90.90	99.85	99.55	96.35
44	90.75	99.65	99.55	97.25

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Table 18: SmallResNet64 Tiny-ImageNet-200 – Per-seed results ($n=3$).

Seed	Glob. Acc	Glob. Conf	Glob. Probe	SC Acc	SC Probe
42	60.88	53.78	51.55	0.00	100.00
43	61.08	54.04	52.60	0.50	100.00
44	60.36	53.44	51.35	0.00	100.00

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